

Assignment No. 5

Bastion Host Setup

Objective:

To implement a secure Bastion Host architecture in AWS that allows access to private EC2 instances without exposing them directly to the internet. SSH Agent Forwarding was used to securely access the private instance through the bastion host.

Services Used:

- Amazon EC2
 - Amazon VPC
 - Security Groups
 - SSH Agent Forwarding
 - Ubuntu Linux
-

Architecture Overview

Laptop > SSH Bastion Host (Public Subnet) > SSH Agent Forwarding Private EC2 (Private Subnet)

Step 1: Create VPC and Subnets

A custom VPC named assignment-5-vpc was created with CIDR block 10.0.0.0/16.

Subnets created:

Public Subnet: 10.0.1.0/24

Private Subnet: 10.0.11.0/24

Purpose:

Public subnet hosts internet-accessible resources.

Private subnet hosts internal resources that are not directly reachable from the internet.

vpc-0c657e6d94c1c4705

You successfully created vpc-0c657e6d94c1c4705 / assignment-5-vpc

vpc-0c657e6d94c1c4705 / assignment-5-vpc

Actions

Details

Info

VPC ID

vpc-0c657e6d94c1c4705

DNS resolution

Enabled

Main network ACL

acl-0c6965e024af3aacc

IPv6 CIDR (Network border group)

-

Encryption control ID

-

State

Available

Tenancy

default

Default VPC

No

Network Address Usage metrics

Disabled

Encryption control mode

-

Block Public Access

Off

DHCP option set

dopt-04b6d2ff08f4fb0ea

IPv4 CIDR

10.0.0.0/16

Route 53 Resolver DNS Firewall rule groups

-

DNS hostnames

Disabled

Main route table

rtb-02fd9e0f4d16262fb

IPv6 pool

-

Owner ID

339333478686

You have successfully created 1 subnet: subnet-0e5de9ba090a7845c

Subnets (1/1)

Info

Last updated less than a minute ago

Actions

Create subnet

Find subnets by attribute or tag

Subnet ID : subnet-0e5de9ba090a7845c

Clear filters

Name

Subnet ID

State

VPC

Block Public...

IPv4 CIDR

public-subnet

subnet-0e5de9ba090a7845c

Available

vpc-0c657e6d94c1c4705 | assign...

Off

10.0.1.0/24

subnet-0e5de9ba090a7845c / public-subnet

Details

Flow logs

Route table

Network ACL

CIDR reservations

Sharing

Tags

Subnet ID

subnet-0e5de9ba090a7845c

IPv4 CIDR

10.0.1.0/24

Availability Zone

aps1-az1 (ap-south-1a)

Network ACL

-

Subnet ARN

arn:aws:ec2:ap-south-1:339333478686:subnet/subnet-0e5de9ba090a7845c

Available IPv4 addresses

251

Network border group

ap-south-1

Default subnet

No

State

Available

IPv6 CIDR

-

VPC

vpc-0c657e6d94c1c4705 | assignment-5-vpc

Auto-assign public IPv4 address

No

Block Public Access

Off

IPv6 CIDR association ID

-

Route table

-

Auto-assign IPv6 address

No

You have successfully created 1 subnet: subnet-09c2d6314472b6a76

Subnets (1/1) Info

Last updated less than a minute ago

Actions

Create subnet

Find subnets by attribute or tag

Subnet ID : subnet-09c2d6314472b6a76

Clear filters

< 1 >

☒	Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR
☒	private-subnet	subnet-09c2d6314472b6a76	Available	vpc-0c657e6d94c1c4705 assign...	Off	10.0.11.0/24

subnet-09c2d6314472b6a76 / private-subnet

DetailsFlow logsRoute tableNetwork ACLCIDR reservationsSharingTags

Details

Subnet ID

subnet-09c2d6314472b6a76

IPv4 CIDR

10.0.11.0/24

Availability Zone

aps1-az1 (ap-south-1a)

Network ACL

-

Auto-assign customer-owned IPv4

Subnet ARN

arn:aws:ec2:ap-south-1:3393334786:86:subnet/subnet-09c2d6314472b6a76

Available IPv4 addresses

251

Network border group

ap-south-1

Default subnet

No

State

Available

IPv6 CIDR

-

VPC

vpc-0c657e6d94c1c4705 | assignment-5-vpc

Auto-assign public IPv4 address

No

Block Public Access

Off

IPv6 CIDR association ID

-

Route table

-

Auto-assign IPv6 address

No

IPv4 CIDR reservations

Step 2: Create Internet Gateway and Route Table

An Internet Gateway was created and attached to the VPC.

A public route table was configured with:

Destination: 0.0.0.0/0 **Target:** Internet Gateway

The public subnet was associated with the public route table.

Purpose: Allows resources in the public subnet to access the internet.

igw-06ead772f821f1e86

Internet gateway igw-06ead772f821f1e86 successfully attached to vpc-0c657e6d94c1c4705

igw-06ead772f821f1e86 / Name assignment-5-igw

Details

Internet gateway ID
igw-06ead772f821f1e86

State
Attached

VPC ID
vpc-0c657e6d94c1c4705 | assignment-5-vpc

Owner
339333478686

Tags (1)

Search tags

Key	Value
Name	Name assignment-5-igw

Updated routes for rtb-0430435d2736ef39a / public-route-table successfully

rtb-0430435d2736ef39a / public-route-table

Details

Route table ID
rtb-0430435d2736ef39a

VPC
vpc-0c657e6d94c1c4705 | assignment-5-vpc

Main
No

Owner ID
339333478686

Explicit subnet associations
-

Edge associations
-

Routes

Subnet associations

Edge associations

Route propagation

Tags

Routes (2)

Filter routes

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-06ead772f821f1e86	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Route Table

Step 3: Launch Bastion Host

A Bastion Host EC2 instance was launched in the public subnet.

Configuration:

Ubuntu Server

Public IP Enabled

Security Group: bastion-sg

Inbound Rule: SSH (Port 22) → My Public IP only

Purpose: Acts as a secure entry point into the private network.

The screenshot displays the AWS Management Console interface for the 'Instances' page. At the top, there's a header with 'Instances (1/1)' and 'Info'. Below this, a table lists the instance 'bastion-host' with ID 'i-04e49e4214c007a2a', state 'Running', type 't3.micro', and availability zone 'ap-south-1a'. The instance is shown as 'Initializing' in the status check column. Below the table, the details for 'i-04e49e4214c007a2a (bastion-host)' are shown. The details are organized into sections: Instance ID, IPv6 address, Hostname type, Answer private resource DNS name, Auto-assigned IP address, Public IPv4 address (13.235.135.95), Instance state (Running), Private IP DNS name (IPv4 only) (ip-10-0-1-146.ap-south-1.compute.internal), Instance type (t3.micro), VPC ID (vpc-0c657e6d94c1c4705), Private IPv4 addresses (10.0.1.146), Public DNS, Elastic IP addresses, and AWS Compute Optimizer finding (Opt-in to AWS Compute Optimizer for recommendations).

The screenshot displays the AWS Management Console interface for the 'Instances' page, specifically the 'Security' tab for the instance 'i-04e49e4214c007a2a (bastion-host)'. The 'Security' tab is selected, showing 'Security details' and 'Inbound rules'. Under 'Inbound rules', there is a table with one rule. The rule has the name 'sgr-0b04233de81603415', security group rule ID 'sgr-0b04233de81603415', port range '22', protocol 'TCP', source '106.210.152.166/32', and security groups 'bastion-sg'. The table is filtered by 'Filter rules'.

Step 4: Launch Private EC2

A private EC2 instance was launched inside the private subnet.

Configuration:

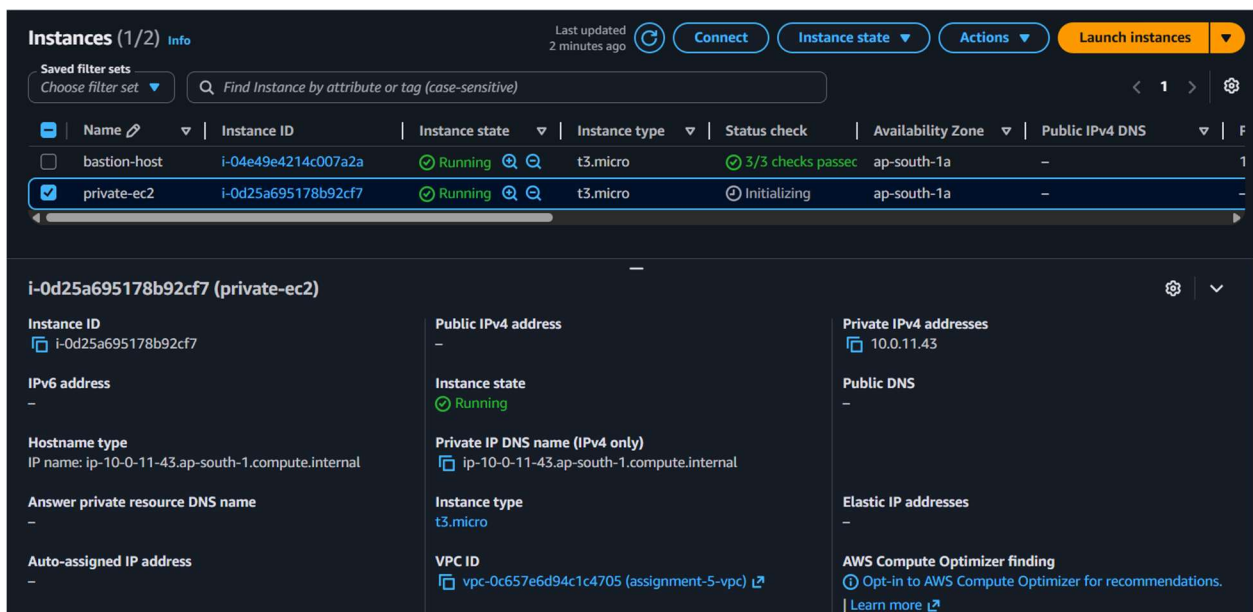
Ubuntu Server

No Public IP

Security Group: private-ec2-sg

Inbound Rule: SSH (Port 22) → bastion-sg

Purpose: Only the Bastion Host can access the private EC2.



Instances (1/2) Info

Last updated 2 minutes ago

Connect Instance state Actions Launch instances

Saved filter sets

Choose filter set Find Instance by attribute or tag (case-sensitive)

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS
bastion-host	i-04e49e4214c007a2a	Running	t3.micro	3/3 checks passed	ap-south-1a	-
private-ec2	i-0d25a695178b92cf7	Running	t3.micro	Initializing	ap-south-1a	-

i-0d25a695178b92cf7 (private-ec2)

Instance ID: i-0d25a695178b92cf7

IPv6 address: -

Hostname type: IP name: ip-10-0-11-43.ap-south-1.compute.internal

Answer private resource DNS name: -

Auto-assigned IP address: -

Public IPv4 address: -

Instance state: Running

Private IP DNS name (IPv4 only): ip-10-0-11-43.ap-south-1.compute.internal

Instance type: t3.micro

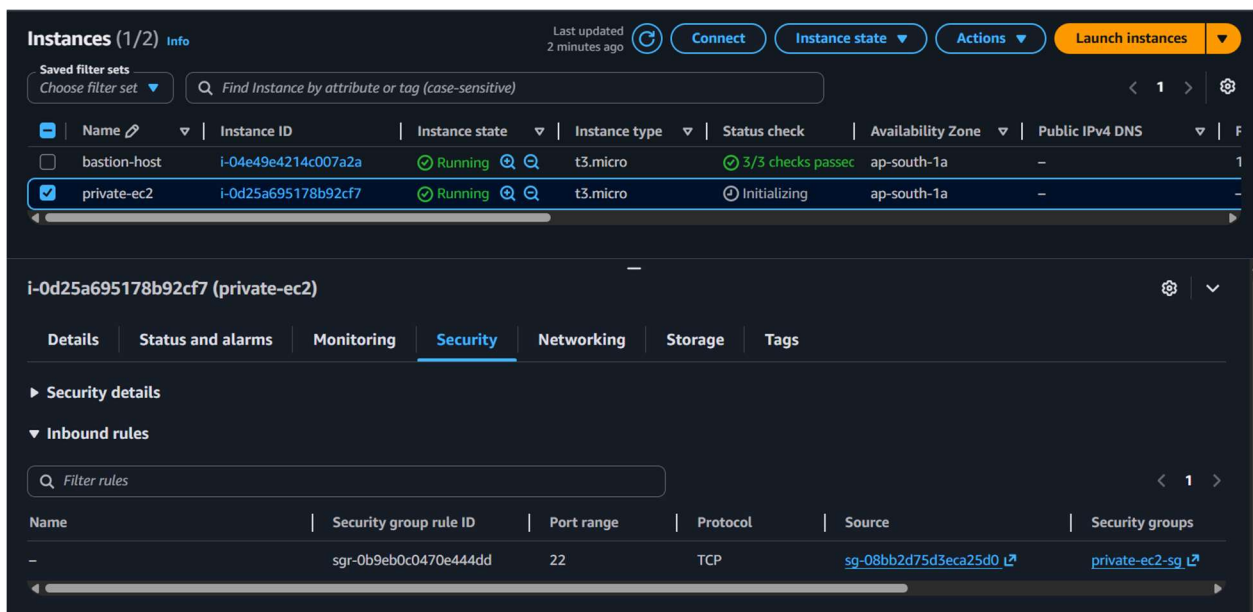
VPC ID: vpc-0c657e6d94c1c4705 (assignment-5-vpc)

Private IPv4 addresses: 10.0.11.43

Public DNS: -

Elastic IP addresses: -

AWS Compute Optimizer finding: Opt-in to AWS Compute Optimizer for recommendations. Learn more



Instances (1/2) Info

Last updated 2 minutes ago

Connect Instance state Actions Launch instances

Saved filter sets

Choose filter set Find Instance by attribute or tag (case-sensitive)

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS
bastion-host	i-04e49e4214c007a2a	Running	t3.micro	3/3 checks passed	ap-south-1a	-
private-ec2	i-0d25a695178b92cf7	Running	t3.micro	Initializing	ap-south-1a	-

i-0d25a695178b92cf7 (private-ec2)

Details Status and alarms Monitoring Security Networking Storage Tags

Security details

Inbound rules

Filter rules

Name	Security group rule ID	Port range	Protocol	Source	Security groups
-	sgr-0b9eb0c0470e444dd	22	TCP	sg-08bb2d75d3eca25d0	private-ec2-sg

Step 5: Configure SSH Agent Forwarding

An SSH configuration file was created on the local system.

The configuration enabled:

Bastion Host access

ProxyJump

Agent Forwarding

Allows SSH authentication through the Bastion Host without copying private keys to AWS instances.

```
PS C:\Users\YASH\.ssh> type $HOME\.ssh\config
Host bastion
    HostName 13.235.135.95
    User ubuntu
    IdentityFile C:\Users\YASH\Downloads\assignment1-key.pem
    ForwardAgent yes

Host private
    HostName 10.0.11.43
    User ubuntu
    ProxyJump bastion
    ForwardAgent yes
PS C:\Users\YASH\.ssh> 
```

Step 6: Connect Through Bastion Host

Connected to Bastion Host using: ssh bastion

Purpose: Verify secure access to the public jump server.

```
PS C:\Users\YASH\.ssh> ssh bastion
The authenticity of host '13.235.135.95 (13.235.135.95)' can't be established.
ED25519 key fingerprint is SHA256:M0zTAlKqJaPtYNYGDG8BfnKGTvZyZdw010d5tkgIVQ.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '13.235.135.95' (ED25519) to the list of known hosts.
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1004-aws x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Mon Jun  1 14:08:47 UTC 2026

System load:  0.08           Temperature:   -273.1 C
Usage of /:   30.4% of 6.61GB Processes:    115
Memory usage: 26%           Users logged in: 0
Swap usage:   0%            IPv4 address for ens5: 10.0.1.146

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

ubuntu@ip-10-0-1-146:~$ |
```

Step 7: Access Private EC2 Through Bastion

Connected to private EC2 using: `ssh ubuntu@10.0.11.43`

Purpose: Verify secure access to the private EC2 through the Bastion Host using SSH Agent Forwarding.

```
ubuntu@ip-10-0-1-146:~$ ssh ubuntu@10.0.11.43
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1004-aws x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Mon Jun  1 14:16:15 UTC 2026

System load:  0.0           Temperature:      -273.1 C
Usage of /:   30.4% of 6.61GB Processes:        114
Memory usage: 26%          Users logged in:  0
Swap usage:   0%           IPv4 address for ens5: 10.0.11.43

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
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applicable law.

ubuntu@ip-10-0-11-43:~$
```

Learning Outcomes

Bastion Host Architecture

Public and Private Subnet Design

Security Group Referencing

SSH Agent Forwarding

AWS Networking Fundamentals

Secure Administrative Access Patterns

Conclusion:

Successfully implemented a Bastion Host architecture in AWS. The private EC2 instance remained inaccessible from the internet while allowing secure administration through the Bastion Host using SSH Agent Forwarding.